

SECTION 2

TARPED MAGNET DELIVERY

2-1 IN-TRANSIT SERVICE

Description:

The following information outlines provisions for filling the magnet with cryogens and performing electrical / temperature checks while magnet the is still in shipping tarp. This will enable required service to be performed in transit based upon magnet shipping date.



Review guidelines with Shipper / Riggers prior to transporting magnet.

Shipping and handling guidelines are provided in Tables 2-1, 2-2 and 2-3.

These guidelines must be followed to prevent any potential damage to the magnet during shipping and handling.

2-1 IN-TRANSIT SERVICE (continued)

LIFTING / HANDLING INFORMATION
TABLE 2-1

MAGNET	* MAXIMUM WEIGHT	** MAXIMUM TILT	FORKLIFT CAPABILITY
CX 1.5T	11,000 lbs. (4,990 kg.)	30 deg.	Yes
CX 1.0T	10,200 lbs. (4,627 kg.)		



The magnet **CANNOT** be shipped by train as serious damage to the magnet's internal suspension system may occur due to the vibration loads introduced by rail systems.

SHIPPING INFORMATION
TABLE 2-2

MAGNET	ALLOWABLE SHIPPING MODES	SHIPPING CAPABILITY	MAXIMUM TRANSIT TIME	COMMENTS
CX 1.0T and 1.5T	Airplane, Air ride only trailer, Boat or ocean going ship	Warm or Cold	Cryogen refill for cold shipment = xx days	See Note for Tie downs

MAXIMUM LOAD INFORMATION
TABLE 2-3

MAGNET	MAXIMUM LONGITUDINAL LOAD	MAXIMUM LATERAL LOAD	MAXIMUM VERTICAL LOAD	MAXIMUM SHOCK LOADS
CX 1.0T and 1.5T	1.2 G	1.5 G	1.5 G	None Allowed

* Includes weight of BRM Coil – 2,300 lbs. (1,044 kg.) and associated hardware – 126 lbs. (58 kg.) including cryogenics. Does NOT include Shipping Pallet – 400 lbs. (182 kg.)

** Maximum tilt allowed only when supported by feet for forklift / rigging operations or moving on inclines (base lift).
Tilt allowed when lifted by single ended slings is one deg. (top lift).

Note:

1. Tie downs and lifting points limited per instructions on magnet face.

2-1 IN-TRANSIT SERVICE (continued)**2-1-1 Helium Fill****WARNING!**

THIS PROCEDURE MUST BE PERFORMED BY QUALIFIED PERSONNEL ONLY. MAKE SURE THE SAFETY PRECAUTIONS LISTED BELOW ARE COMPLETELY FOLLOWED WHEN PERFORMING ANY SERVICE THAT WILL RESULT IN HELIUM EXHAUST FROM THE MAGNET.

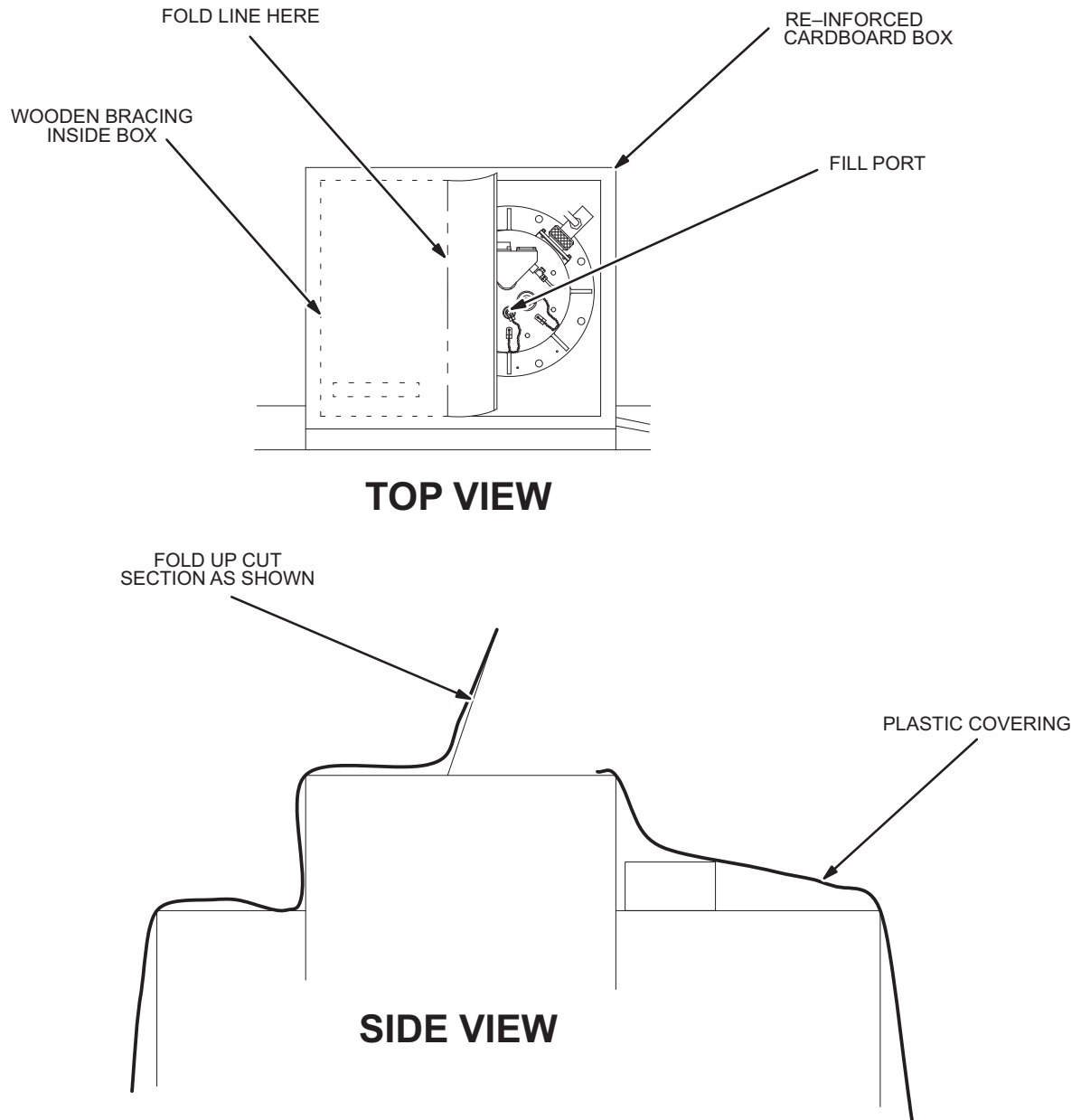
LARGE QUANTITIES OF HELIUM GAS WILL DISPLACE OXYGEN IN A CLOSED ROOM. FOR YOUR PERSONAL SAFETY, MAKE SURE THE MAGNET STORAGE AREA IS ADEQUATELY VENTILATED BY OPENING DOORS AND WINDOWS. THE ADDITION OF PORTABLE FANS MAY BE NECESSARY IN CASES OF SMALL ROOM SIZE WITH POOR VENTILATION. THIS IS REQUIRED.

SKIN CONTACT WITH LIQUID CRYOGENS WILL CAUSE BURNS. WEAR PROTECTIVE CLOTHING, GLOVES (NONABSORBENT MATERIAL) AND GOGGLES OR FACE SHIELD WHEN TRANSFERRING CRYOGENS. MAKE SURE A SECOND PERSON IS PRESENT WHILE SERVICING THE MAGNET IN CASE OF ANY EMERGENCY.

SMOKING IS PROHIBITED IN THE MAGNET AND CRYOGEN STORAGE ROOMS. LIQUID CRYOGENS CAN LIQUIFY ATMOSPHERIC OXYGEN THUS PRODUCING A HIGHLY ENRICHED OXYGEN LIQUID.

2-1-1 Helium Fill (continued)**NOTE:**

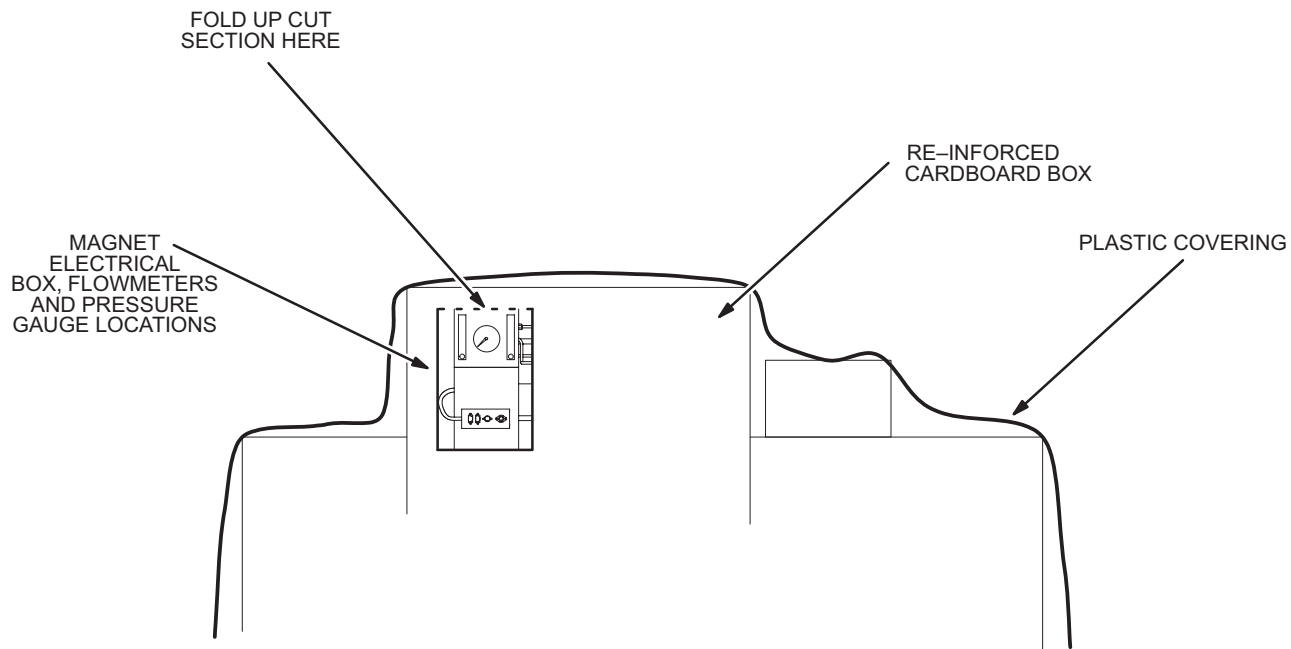
TO REFILL THE MAGNET BEFORE IT'S ARRIVAL AT SITE, CARE MUST BE TAKEN TO CUT THE CARDBOARD BOX AND PLASTIC COVERING ACCESS TO THE MAGNET FILL PORT. CUTS IN THE BOX AND PLASTIC MUST BE TAPED CLOSED ADEQUATELY ENOUGH TO SEAL OUT DIRT AND MOISTURE AFTER THE FILL PROCEDURE IS COMPLETE.



HELIUM FILL ACCESS
ILLUSTRATION 2-1

2-1-2 Electrical Checks, Flowmeter and Pressure Gauge Checks**NOTE**

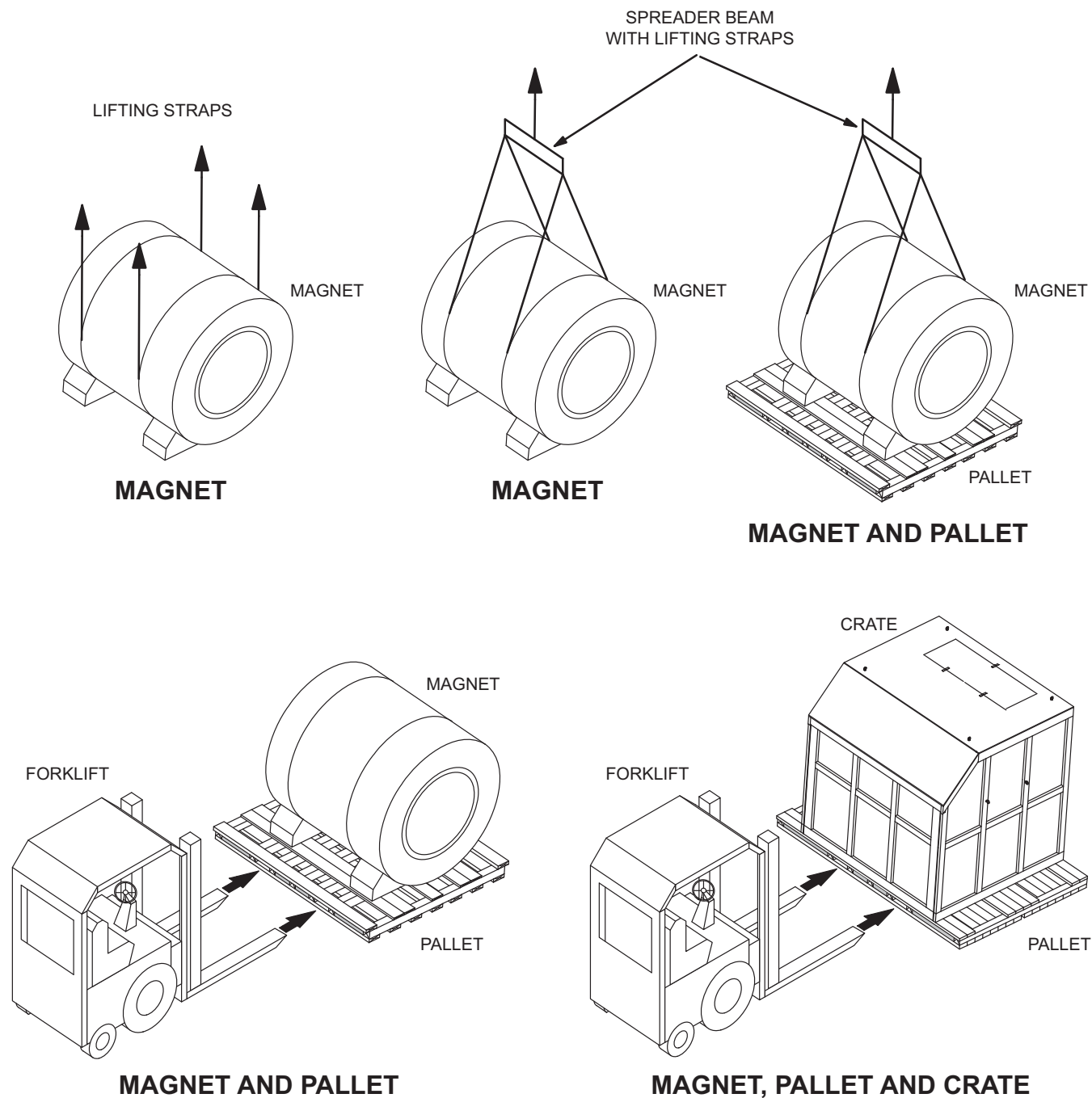
CARDBOARD BOX AND PLASTIC COVERING SHOULD BE CUT NO MORE THAN NECESSARY. CUTS IN THE BOX AND PLASTIC MUST BE TAPED CLOSED ADEQUATELY ENOUGH TO SEAL OUT DIRT AND MOISTURE AFTER CHECKS ARE COMPLETE.



ELECTRICAL, FLOWMETER AND PRESSURE GAUGE CHECKS
ILLUSTRATION 2-2

2-2 TARP AND PROTECTIVE PACKAGING REMOVAL

Magnet is shipped in wrapped plastic. Care must be taken in removing the plastic to avoid damage to various magnet components.

2-3 UNLOADING MAGNET AND PALLET**2-3-1 ACCEPTABLE LIFTING CONFIGURATIONS FOR Cx ACTIVE SHIELD MAGNET, PALLET AND CRATE**

LIFTING CONFIGURATIONS
ILLUSTRATION 2-3

2-3-2 Forklift Method**CAUTION**

It is important to use the proper size forklift and to lift the magnet smoothly to avoid impact or jolts to the system which may cause damage to the magnet. Extreme care must be exercised during forklift operations. The magnet pallet must be picked up from the sides only. The forks **MUST** be placed under all four (4) feet of the magnet. Make sure the forklift capacity is rated for the weight of the crate and the contents. See Illustration 2-4 for required fork dimensions. The minimum fork length is 2.1 Meters (80 inches).

1. Locate the forklift openings on the side of the shipping pallet. See Illustration 2-4.
2. Extreme care must be exercised during forklift operations. The magnet / pallet must be picked up from the sides only. The forks **MUST** be placed directly under the four (4) feet of the magnet. Position the forks at the shipping pallet forklift openings. See Illustration 2-4.
3. Carefully drive the forklift until forks are completely inserted into the pallet openings.
4. Lift magnet / pallet with forklift and move smoothly to desired location.

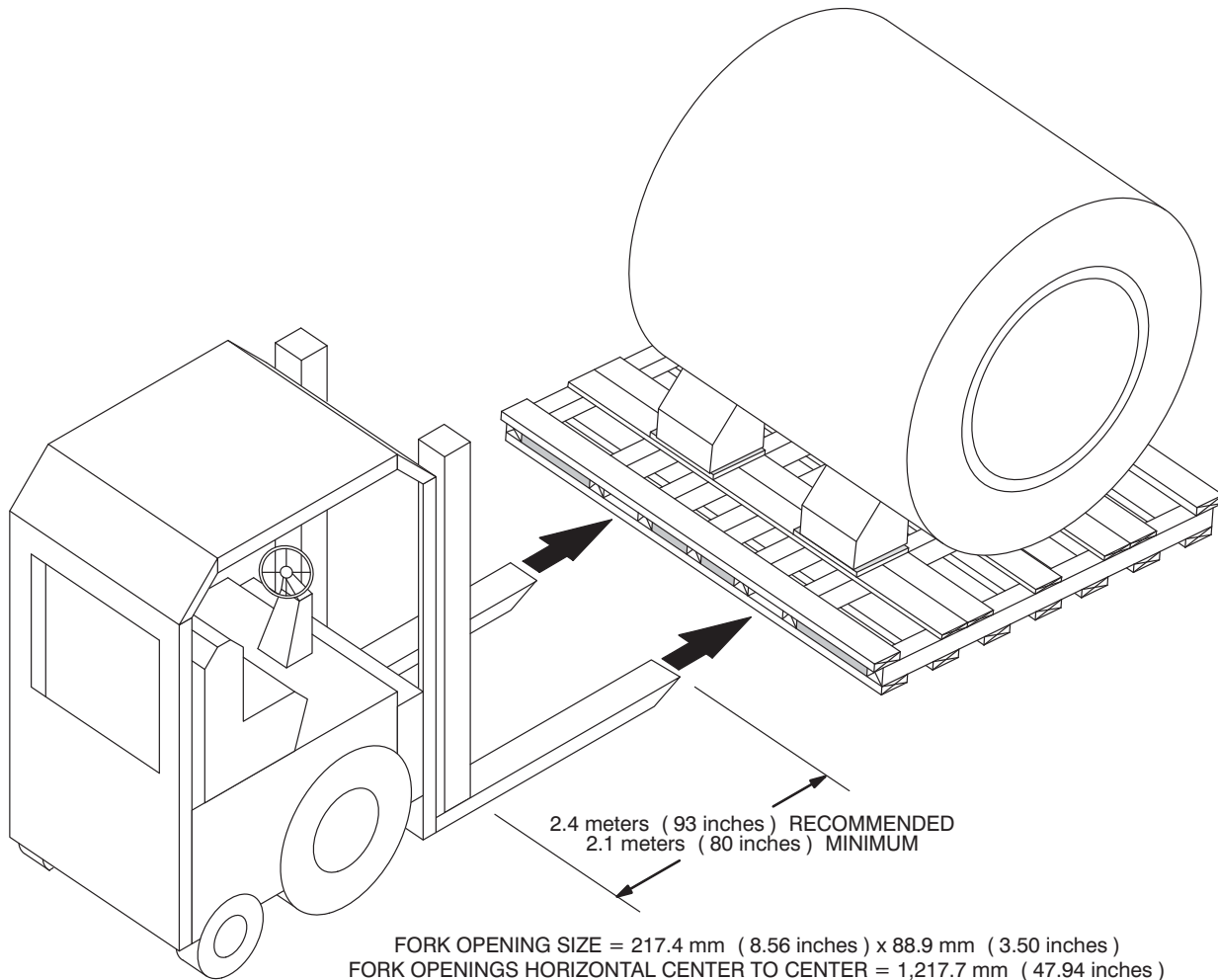
**CRATE / PALLET FORKLIFT OPENINGS**

ILLUSTRATION 2-4

2-3-3 Crane Method

The lifting hooks of the spreader beam **MUST** be located at the dimensions shown in Illustration 2-6. This dimension is based on 3.00" (76.2 mm) wide lifting straps rated at 11,000 lbs. (4,990 kg.) minimum.

Lifting straps must be long enough to allow a minimum of 24.00" (609.6 mm) clearance between the top of the magnet outer vacuum vessel and the top of the lifting strap. See Illustrations 2-6 and 2-7.

Pallet can be lifted by magnet. Pallet does not need to be unbolted from magnet prior to lifting operation.

Failure to follow these requirements can result in serious damage to the magnet.

2-3-3 Crane Method (continued)

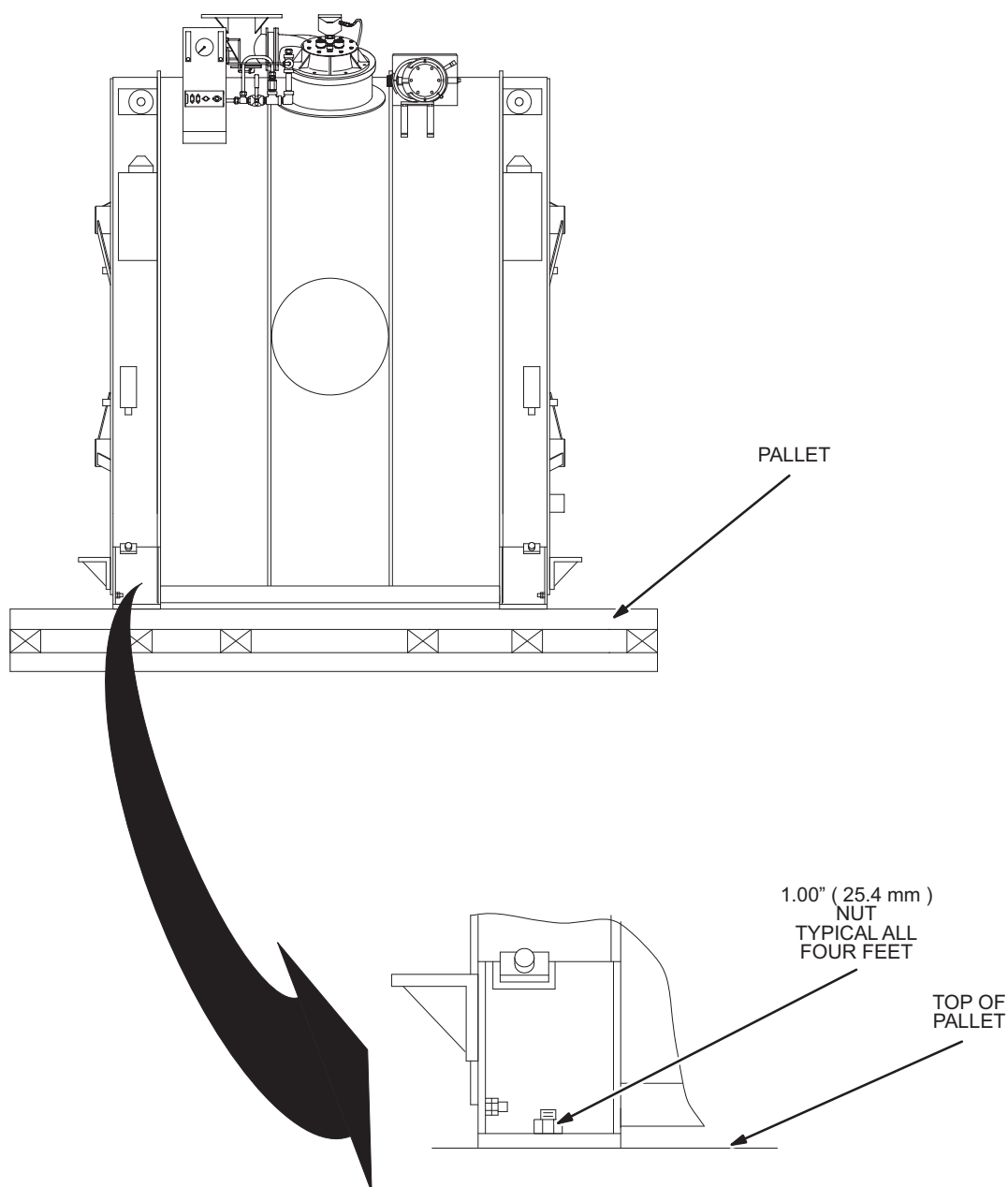
It is important to lift the magnet smoothly to avoid impact or jolts to the system. If severe jolting is encountered, internal damage may result. The magnet cannot be tipped by more than 1 degree during crane lifting operation.

1. Unbolt and remove the four 1 inch (25.4 mm) nuts which secure the pallet to the magnet. See Illustration 2-5.
2. Position the crane hook centrally over the magnet to ensure a vertical lift force on the lifting straps. See Illustration 2-6 and 2-7.
3. Attach the rigging to the lifting straps at each end of the magnet. See Illustration 2-6 and 2-7.
4. Lift magnet with crane to clear shipping pallet.

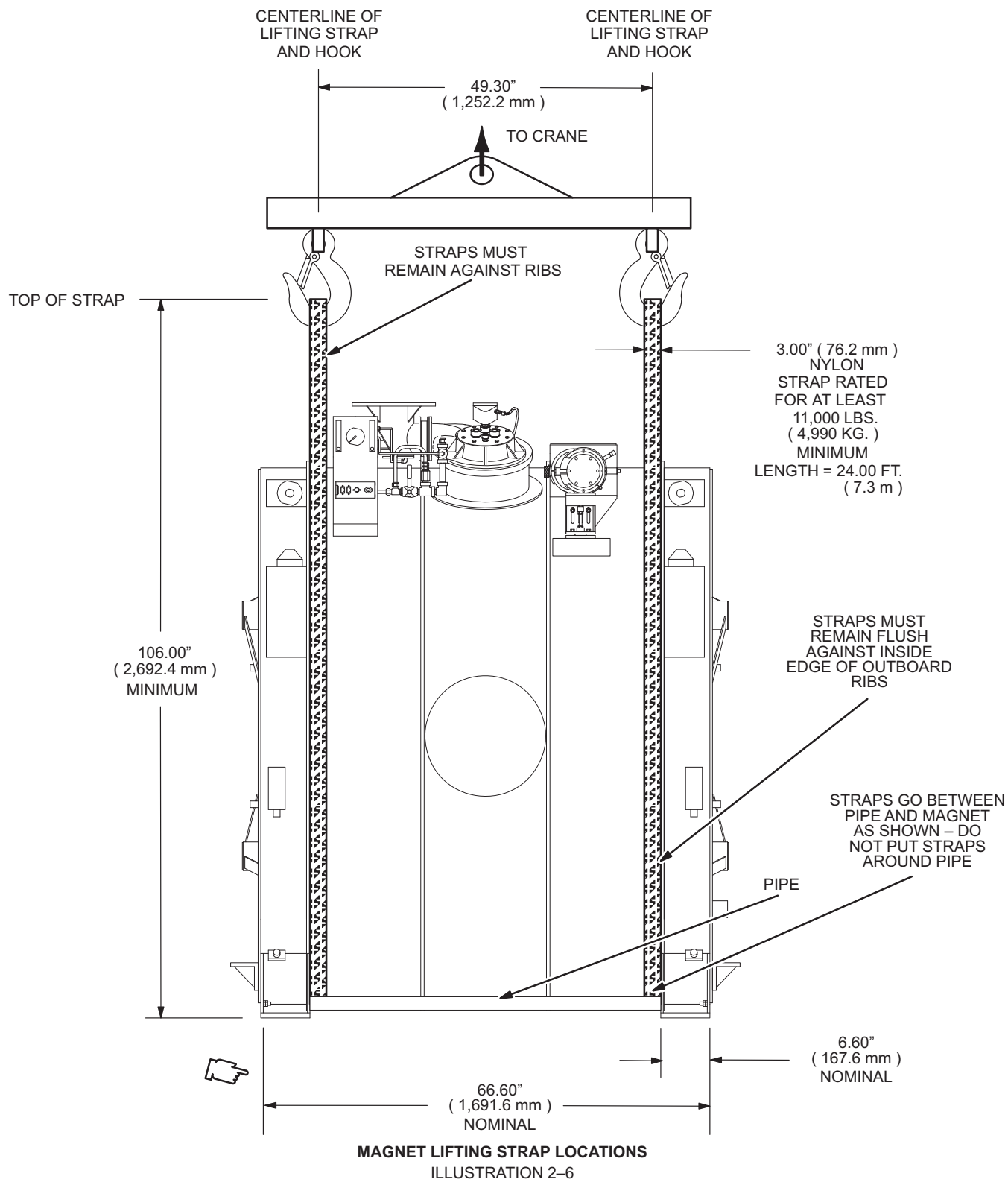


When lowering the magnet, be careful not to shock the magnet which could result in over stressing the internal magnet support members.

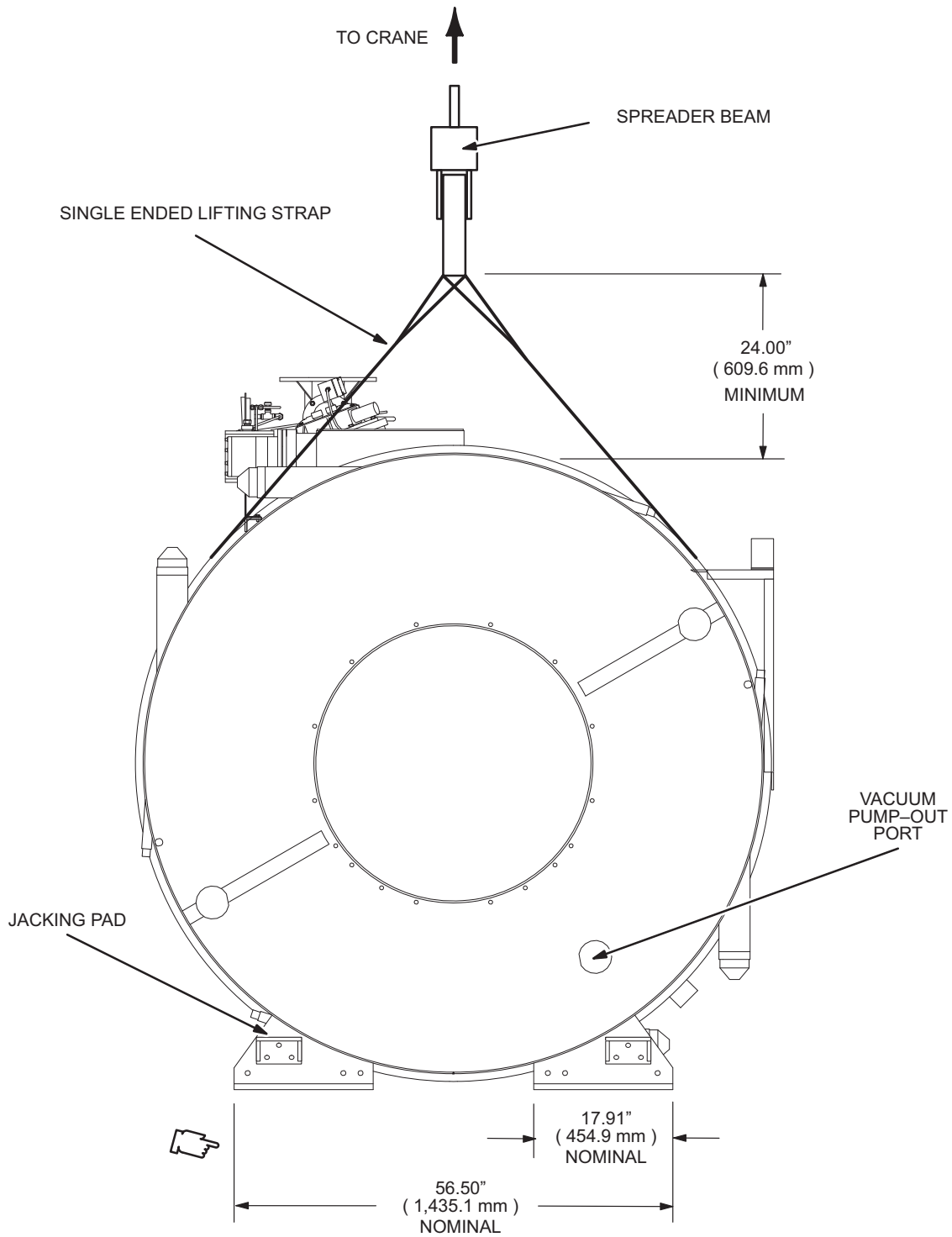
5. Lower the magnet onto a flat surface.

2-3-3 Crane Method (continued)

MAGNET MOUNTING ON SHIPPING PALLET
ILLUSTRATION 2-5

2-3-3 Crane Method (continued)

2-3-3 Crane Method (continued)



MAGNET LIFTING STRAP HEIGHT
ILLUSTRATION 2-7

